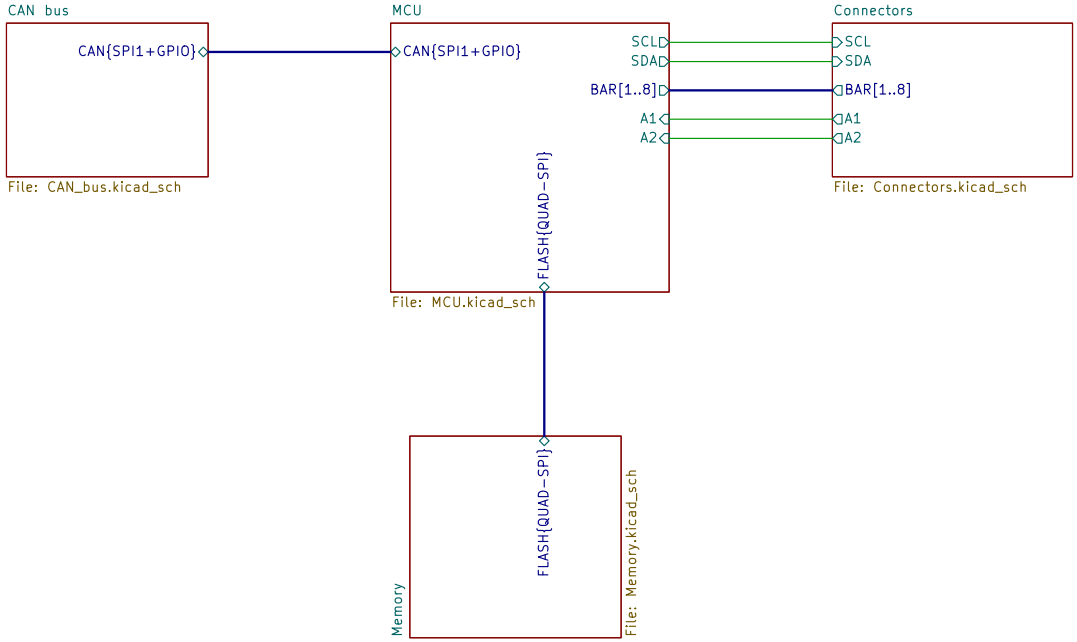


H101
MountingHole

H102
MountingHole

H103
MountingHole

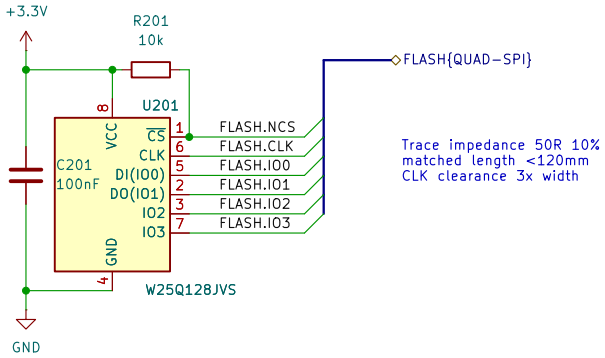
RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute

Sheet: /
File: BB_tunnel_V1.kicad_sch

Title: BB_tunnel

Size: A4	Date: 2024-03-21	**Rev: v1**
KiCad E.D.A. 10.0.1		Id: 1/5

Quad-SPI FLASH NOR Memory

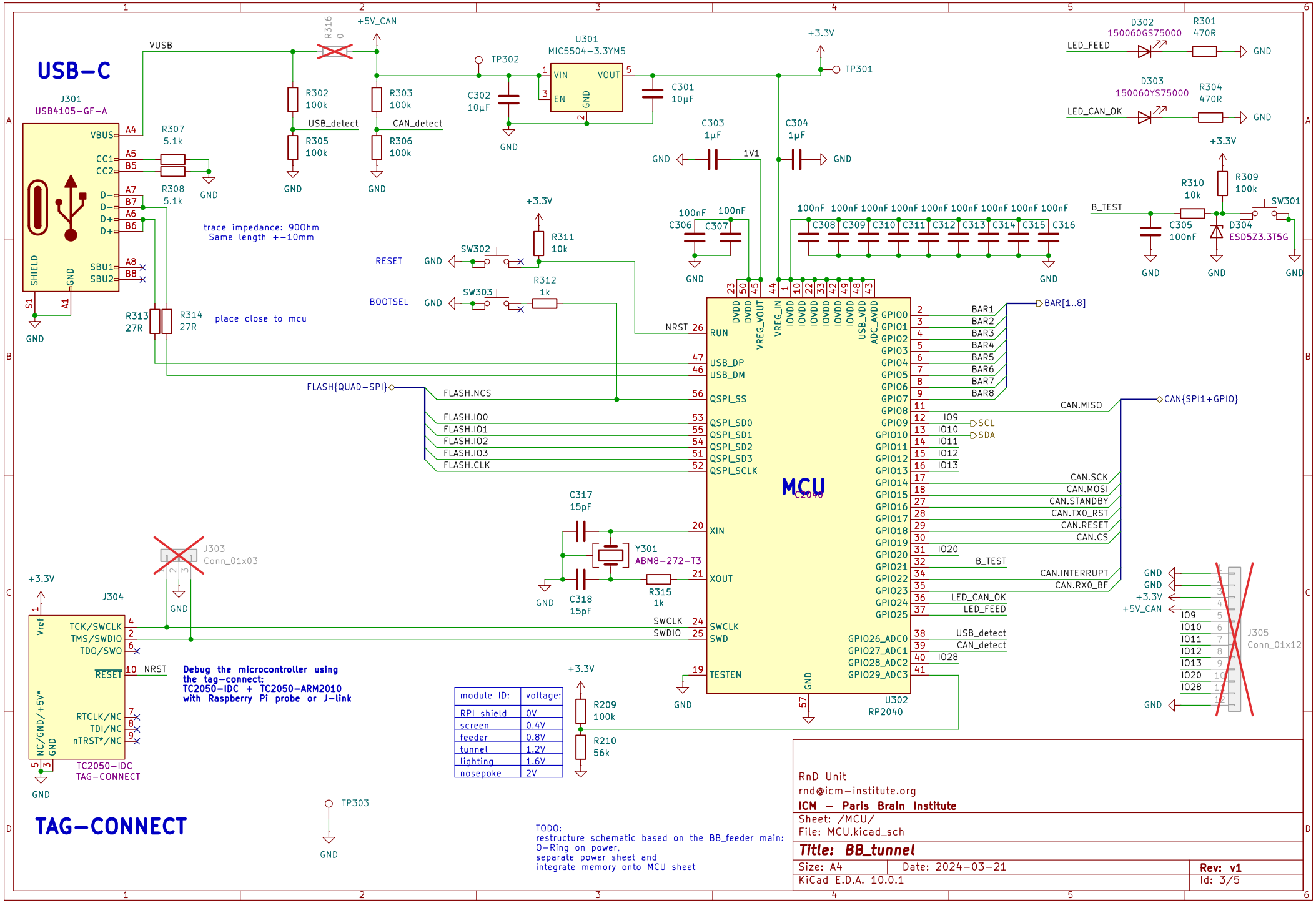


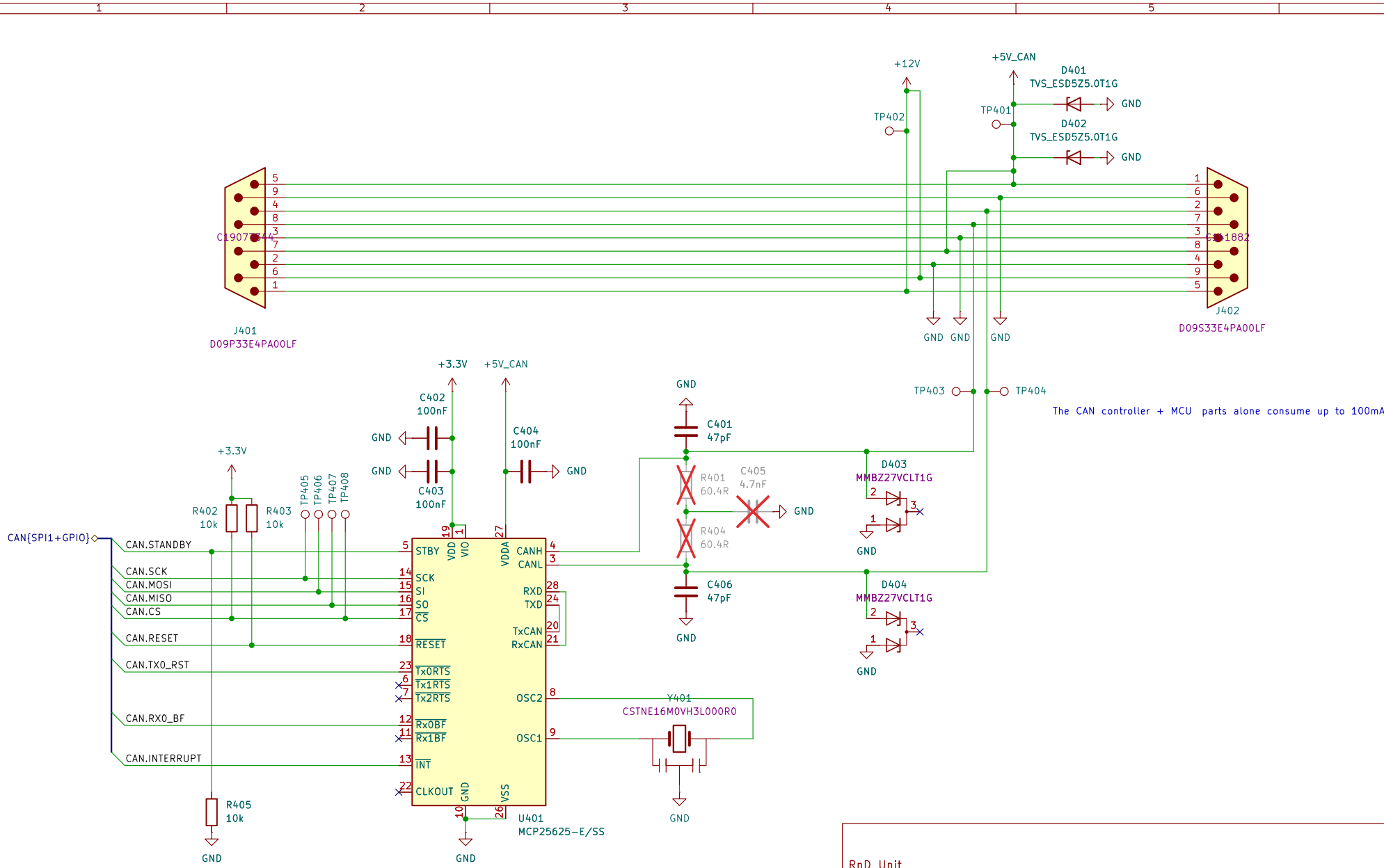
RnD Unit
rnd@icm-institute.org
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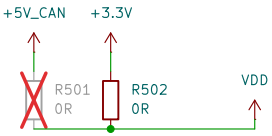
Sheet: /Memory/
File: Memory.kicad_sch

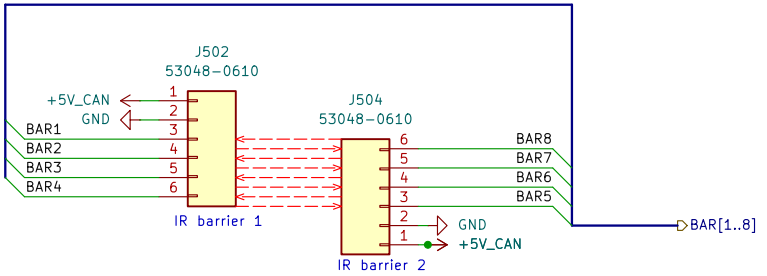
Title: BB_tunnel

Size: A4	Date: 2024-03-21	**Rev: v1**
KiCad E.D.A. 10.0.1		Id: 2/5

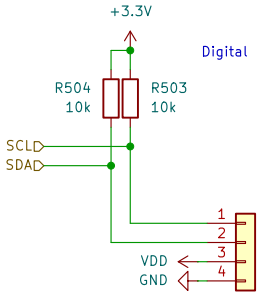

USB-C
J301
USB4105-GF-A
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
S1
A1
VUSB
R316
0
+5V_CAN
R307
5.1k
R308
5.1k
R302
100k
R303
100k
R305
100k
R306
100k
USB_detect
CAN_detect
trace impedance: 90Ohm
Same length +-10mm
U301
MIC5504-3.3YM5
VIN
VOUT
EN
GND
TP302
C302
10µF
C301
10µF
+3.3V
TP301
C303
1µF
1V1
C304
1µF
D302
150060GS75000
R301
470R
LED_FEED
D303
150060YS75000
R304
470R
LED_CAN_OK
R310
10k
R309
100k
B_TEST
SW301
C305
100nF
D304
ESD5Z3.3T5G
C306
C307
100nF
C308
C309
C310
C311
C312
C313
C314
C315
C316
R311
10k
SW302
RESET
SW303
BOOTSEL
R312
1k
R313
27R
R314
27R
place close to mcu
MCU
RP2040
NRST
RUN
USB_DP
USB_DM
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
XIN
XOUT
SWCLK
SWD
TESTEN
FLASH{QUAD-SPI}
FLASH.NCS
FLASH.IO0
FLASH.IO1
FLASH.IO2
FLASH.IO3
FLASH.CLK
C317
15pF
C318
15pF
Y301
ABM8-272-T3
R315
1k
J303
Conn_01x03
J304
TC2050-IDC
TAG-CONNECT
TCK/SWCLK
TMS/SWDIO
TDO/SWO
RESET
NRST
RTCLK/NC
TDI/NC
nTRST*/NC
Debug the microcontroller using
the tag-connect:
TC2050-IDC + TC2050-ARM2010
with Raspberry Pi probe or J-link
R209
100k
R210
56k
module ID:
voltage:
RPI shield
0V
screen
0.4V
feeder
0.8V
tunnel
1.2V
lighting
1.6V
nosepoke
2V
TP303
TAG-CONNECT
TODO:
restructure schematic based on the BB_feeder main:
O-Ring on power,
separate power sheet and
integrate memory onto MCU sheet
BAR[1..8]
CAN{SPI1+GPIO}
SCL
SDA
CAN.MISO
CAN.SCK
CAN.MOSI
CAN.STANDBY
CAN.TX0_RST
CAN.RESET
CAN.CS
CAN.INTERRUPT
CAN.RX0_BF
LED_CAN_OK
LED_FEED
USB_detect
CAN_detect
B_TEST
J305
Conn_01x12
RnD Unit
rnd@icm-institute.org
ICM - Paris Brain Institute
Sheet: /MCU/
File: MCU.kicad_sch
Title: BB_tunnel
Size: A4
Date: 2024-03-21
KiCad E.D.A. 10.0.1
Rev: v1
Id: 3/5


+12V
+5V_CAN
D401
TVS_ESD5Z5.0T1G
GND
TP401
D402
TVS_ESD5Z5.0T1G
GND
TP402
C190744
J401
D09P33E4PA00LF
C191882
J402
D09S33E4PA00LF
GND GND GND
TP403
TP404
The CAN controller + MCU parts alone consume up to 100mA
+3.3V
+5V_CAN
C402
100nF
GND
C404
100nF
GND
GND
C403
100nF
+3.3V
R402
10k
R403
10k
TP405
TP406
TP407
TP408
CAN{SPI1+GPIO}
CAN.STANDBY
CAN.SCK
CAN.MOSI
CAN.MISO
CAN.CS
CAN.RESET
CAN.TX0_RST
CAN.RX0_BF
CAN.INTERRUPT
STBY
SCK
SI
SO
CS
RESET
TX0RTS
TX1RTS
TX2RTS
RX0BF
RX1BF
INT
CLKOUT
VDD
VIO
VDDA
GND
VSS
CANH
CANL
RXD
TXD
TxCAN
RxCAN
OSC2
OSC1
U401
MCP25625-E/SS
R405
10k
GND
GND
C401
47pF
R401
60.4R
C405
4.7nF
GND
R404
60.4R
C406
47pF
GND
D403
MMBZ27VCLT1G
GND
D404
MMBZ27VCLT1G
GND
Y401
CSTNE16M0VH3L000R0
GND
RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute
Sheet: /CAN bus/
File: CAN_bus.kicad_sch
Title: BB_tunnel
Size: A4
Date: 2024-03-21
Rev: v1
KiCad E.D.A. 10.0.1
Id: 4/5





The mouse will trigger BAR1 first and BAR5 last



STEMMA QT / Qwiic
Can be used with other compatible modules (STEMMA, Gravity, GROVE) with the right adapter cable.
More info about compatibility here: https://learn.adafruit.com/introducing-adafruit-stemma-qt/stemma-qt-comparison

RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute

Sheet: /Connectors/
File: Connectors.kicad_sch

Title: BB_tunnel

Size: A4	Date: 2024-03-21	**Rev: v1**
KiCad E.D.A. 10.0.1		Id: 5/5